

Phase 1 Navigation — Project Plan V2

Sowbot × iFarmate / APSI collaboration ·

Contracting parties: the engagement is contracted with [AgroecologyLab](#) (Sowbot), which provides the platform, mentoring, code review and the hardware items noted as Sowbot-supplied below. [APSI](#) / iFarmate provide the two developers and the i-Farmate chassis.

1. Objective

Phase 1 delivers autonomous in-field navigation only — no agricultural attachments. The success criterion is a single capability demonstrated repeatably in a real maize field:

- **Row traversal:** the robot autonomously drives to the end of a crop row using visual row-following.
- **Headland turn:** on reaching the row end it transitions to the next adjacent row and resumes traversal.
- **Repeat:** the cycle continues across a short block of rows under a single mission command, with safe stop on E-stop / heartbeat loss.

Weeding and seeding attachments are explicitly deferred to Phase 2.

2. Technical approach

Phase 1 reuses the existing two-board architecture already scaffolded in `feldfreund_devkit_ros`. Work divides cleanly along that hardware boundary, which is why the two developers can own one board each with minimal day-to-day coupling — the contract between them is a small set of ROS 2 topics and one action.

Board	Role	Phase 1 responsibility
Neo (Perception)	Vegetation index + row-line detection (<code>sowbot_row_follow</code>)	Detects the crop row, publishes <code>/aoc/conditions/row_offset</code> and <code>row_heading_error</code> plus a vision heartbeat. Enabled on demand over the 1 GbE link. Row-following perception is tested and iterated against the field, not treated as fixed.

Board	Role	Phase 1 responsibility
Limbic (Executive)	Topological nav + Nav2 + GNSS fusion	Runs the mission state machine and the /limbic_row_follow NavigateToPose action: visual servo to row end, then Nav2 hand-off for the headland turn and heading alignment.
Lizard (RT MCU)	Motor PID + safety	Accepts v , ω over the teleop bridge. On the i-Farmate body this layer is brought up against the chosen motor driver (see §5) using the Lizard RoboClawWheels or ODriveWheels module.

The handover contract is the integration backbone for the two roles:

- **Neo** → **Limbic**: row offset, row heading error, and a vision heartbeat on the Zenoh / GbE link.
- **Limbic** → **Neo**: a SetBool enable on /row_follow/enable so perception only runs when a row-traversal segment is active.
- **Safety**: heartbeat loss or action cancel disables Neo and aborts cleanly before any further motion.

On perception being iterative: the crop_row_node controller, topic structure and service interface stay stable, but the perception that feeds them — the vegetation index, masking, contour and scan-window tuning — is expected to be measured against real maize and revised each field session. The plan budgets explicit iteration time for this rather than assuming the first pipeline is final.

3. Team & ownership

Two developers, one board each. This mirrors the codebase split and keeps the weekly PR streams independent. The table below is the combined ownership view across both roles.

	Neo — Perception - Subathra & Seenu	Limbic — Nav New developer with ROS experience
Owns	sowbot_row_follow on the Neo board: camera, ExG/Otsu vegetation index, scan-window line fit, visual-servo conditions — tested and iterated against the field.	Limbic executive: topological_navigation (AOC), Nav2, fusioncore GNSS fusion, the /limbic_row_follow action server.
Initial hardware	1× Raspberry Pi camera (CSI), plus 1× USB camera as fallback.	3× u-blox F9P: moving-base + rover for dual-antenna heading, plus a static base.
Primary skills	Python, OpenCV, basic ROS 2 pub/sub & QoS.	Python + ROS 2 (Nav2, lifecycle, actions, TF), GNSS/RTK familiarity.
Definition of done	Conditions published at stable rate with correct QoS; heartbeat reliable; pipeline iterated and calibrated to the maize plot.	Action server completes row + headland segments in sim, then on hardware; fused odom stable.

Working cadence (per the meeting agenda):

- Minimum one pull request per developer per week.
- 1–2 tickets allocated per developer per week; on completion they pull the next open ticket.
- Weekly ~1 hr sync; ~2 hr review & merge; ~2 hr core dev time on the Sowbot side.
- Invoicing weekly.

4. Milestones & timeline

A four-month plan (mid-June to mid-October 2026), structured in four blocks of roughly four weeks each. Each milestone names an owner — Neo, Limbic, or Both — and a concrete acceptance test. Dates assume a kick-off the week of 15 June 2026. The extra month over a bare-minimum schedule is deliberately spent on the i-Farmate chassis retrofit and on iterating perception against real crop.

Month 1 — Onboarding, chassis & simulation (Weeks 1–4)

Wk	Owner	Milestone	Acceptance
1	Both	Dev environment + build green	Both devs build the caatinga-dev stack in Docker, run sowbot_sim, and merge a first trivial PR each from their forks to prove the workflow.
1–2	Both	i-Farmate chassis retrofit	Survey the i-Farmate drivetrain, power and mounting; evaluate fitting the Sowbot electronics stack (Lizard, Limbic, Neo, camera, F9P). Lizard drives the wheels via RoboClawWheels (or ODriveWheels if the brushless route is taken — see §5). Teleop drive test passes; E-stop cuts motion. Arrange the retrofit internally
2-3	Limbic	Sim row-follow end-to-end	Robot completes a row-traversal segment in Gazebo; one full mission runs in sim without hardware.
2-3	Neo	Neo vision on bench	sowbot_row_follow runs on the Pi camera; row_offset / row_heading_error publish on a potted plant-row target; heartbeat visible.
4	Limbic	F9P bring-up	Single-receiver ublox_single brings up a fix into /gnss/fix at correct QoS; dual-F9P moving-base publishes NAV-RELPOSNED heading; static base streams corrections.

Month 2 — Integration on hardware (Weeks 5–8)

Wk	Owner	Milestone	Acceptance
5	Limbic	GNSS fusion stable	fusioncore fuses fix + heading + wheel odom into /fusion/odom with a continuous odom→base_link TF; no silent GPS dropout over a 10-min static log.
5–6	Neo	Camera calibration + servo tuning	camera_height_m / camera_tilt_deg measured; visual servo holds a straight artificial maize row at low speed on the chassis, when manually manipulated. First perception-iteration cycle.
7	Both	Neo–Limbic handshake on hardware	/row_follow/enable toggles Neo from Limbic over the 1 GbE link; heartbeat-loss and cancel both safely disable Neo and abort.
8	Both	First real-row traversal	Robot drives one real maize row to the end under /limbic_row_follow on a calm day; manual E-stop verified.

Month 3 — Headland turn & perception iteration (Weeks 9–12)

Wk	Owner	Milestone	Acceptance
9	Limbic	Headland turn (Nav2)	At row end the action hands off to Nav2; robot turns and aligns onto the next adjacent row pose in the field. (Can be prototyped in SIM)
10	Both	Two-row cycle	Robot completes row → turn → next row → traverse

Wk	Owner	Milestone	Acceptance
			autonomously under one mission command. (SIM)
11	Neo	Perception robustness iteration	Using logs from weeks 8–10, revise the vegetation index / masking / contour tuning against real maize: handle sparse emergence, weed pressure and mud splash; recover from a single dropped detection without manual intervention.
12	Both	Multi-row block	3–4 consecutive rows traversed reliably with the iterated perception pipeline.

Note on RTK GPS in occluded environments

Fusioncore's UKF keeps propagating from wheel odometry and IMU when GNSS drops out, so short outages bridge without position loss. ZUPT suppresses velocity drift when the robot's stationary, and the Mahalanobis gate rejects noisy multipath fixes rather than letting them corrupt the state estimate. The open question for the Indian suburban context is how long the outages are in practice and whether dead-reckoning drift accumulates to a problematic level across a full run.

Several features need testing and iterating in the target environment: BNO085 IMU fusion is wired but not yet active in fusioncore; `gnss.min_fix_type` may need relaxing from `RTK_FIXED` to accept `FLOAT` fixes where multipath is heavier; NTRIP via cellular and NavIC on the F9P both need populating and validating as ways to improve fix quality and re-convergence. The TMAP2 topological layer already provides some resilience by following pre-surveyed edges rather than live coordinates, but how well dead-reckoning bridges node gaps under local conditions needs characterising.

If that combination still isn't holding adequate waypoint accuracy once properly exercised, [ROVIO](#) visual-inertial odometry is the fallback — but it brings real integration overhead and the preference is to exhaust the existing stack first.

Month 4 — Field hardening & acceptance (Weeks 13–16)

Wk	Owner	Milestone	Acceptance
13	Both	Repeatability runs	The multi-row mission repeated across several sessions / light conditions; failures logged and triaged.
13–14	Neo	Final perception iteration	Close out the row-following tuning against the repeatability-run failures; parameters locked into crop_row_params.yaml per crop stage.
15	Both	Robustness + recovery	Graceful behaviour on perception loss, GPS dropout and E-stop; clean resume after interruption.
16	Both	Phase 1 acceptance demo	Documented field run of a multi-row block for the APSI team; short report + bag/log bundle delivered. Phase 1 sign-off.

Buffer: weeks 11–16 carry the slack — field testing is weather- and crop-stage dependent, and the most likely slip is moving from sim to live maize. If row detection or RTK proves harder than expected, the multi-row target reduces to a reliable single-row traversal plus a demonstrated single headland turn, with the remaining time spent on perception iteration.

5. Hardware required

Combined hardware across both roles. The drivetrain electronics choice on the i-Farmate body is the one open decision — see the note below the table.

Item	Qty	Notes / owner
Avaota A1 SBC (4 GB / 32 GB) or alternate appropriate SBC	2	Do not order these yet, Still under review One per developer — the Limbic and Neo boards (Allwinner

Item	Qty	Notes / owner
		T527). Pine64 / Yuzuki Avaota A1 .
u-blox F9P GNSS receiver	3	Limbic — moving-base + rover for dual-antenna heading, plus a static base station. F9P
GNSS antennas + cabling	3	Limbic — lever-arm baseline ≥ 0.3 m; offset measured before field use. (see F9P above)
Raspberry Pi camera (CSI)	1	Neo — primary row-detection input. USB camera retained as a fallback.
BNO085 9-DOF IMU breakout	1	Orientation/heading into fusioncore via the Lizard ImuBno085 module (UART-RVC or I ² C). Adafruit BNO085 .
RoboClaw 2x15A motor controller	2	Lizard RoboClawWheels path for the i-Farmate brushed/brushed-equivalent drivetrain. RoboClaw 2x15A, 6–34 VDC .
— or ODrive (brushless upgrade)	—	Optional upgrade to brushless via the Lizard ODriveWheels module. Drivetrain choice is the Indian team's call — see decision note below.
CAN PAL transceiver (ODrive route only)	1	Only if the ODrive route is taken — CAN transceiver between the Lizard ESP32 and the ODrive(s). Adafruit CAN PAL . Seek Sowbot advice before committing to the ODrive route.
NTRIP / RTK correction source	1	Account or local base for RTK fix; needed by week 4–5.

Item	Qty	Notes / owner
i-Farmate chassis	1	Retrofit target; drivetrain documentation / contacts needed for week-8 survey.
Maize test plot access	—	Block of rows at a suitable crop stage for the field-work weeks (8–16).

Drivetrain decision (Farmate team): the default is two RoboClaw 2x15A controllers driven by the Lizard RoboClawWheels differential-drive module — simplest path, accepts v , ω directly. These should have encoders, or be retrofitted with encoders. The alternative is brushless ODrive controllers via the Lizard ODriveWheels module, which adds clean encoder odometry and more headroom, accuracy and robustness but more bring-up effort. Either is supported by Lizard with no change to the teleop_lizard interface above it, so the choice can be made on cost/availability without affecting the nav stack. Confirm before procurement in week 1. If the more capable ODrive route is taken, it needs the CAN PAL transceiver above — **seek Sowbot advice first**, as ODrive bring-up via Lizard is slightly more involved.

Sowbot can also supply: if the Indian team does not already have them, Sowbot (Agroecology Lab) can offer a custom ESP32 carrier PCB for the Lizard Brain and clean, filtered power supplies for the SBCs and blank mounting plates. Worth raising in week 1 alongside the drivetrain decision so procurement is settled in one pass.

6. Targeted crop

Maize is the initial target for Phase 1. It is the crop the existing topological map (get_maize_topo.py / maize_map) and the virtual_maize_field simulation are already built around, so sim and field share the same geometry. Row spacing and plant height for the actual plot should be captured early (week 1–2) so the camera mounting height and tilt, and the row-detection scan window, can be set correctly — and revisited at each perception-iteration milestone.

7. Risks & dependencies

Risk	Likelihood	Mitigation
Sim-to-field gap in row detection	Medium	Bench + artificial-row testing in months 1–2 before relying on live crop; perception is iterated

Risk	Likelihood	Mitigation
		against the real plot at weeks 11, 13–14 rather than assumed final.
i-Farmate drivetrain not Lizard-drivable as-is	Medium	Week-1 survey is the decision gate; RoboClaw path is the low-risk default, assuming encoders present/retrofitted, ODrive the upgrade. .
RTK heading instability (lever-arm, multipath)	Medium	Measure antenna baseline precisely; validate static log before motion; single-receiver fallback path exists.
Weed pressure / mud splash degrading ExG	Medium	Maize Phase 1 timing favours low weed pressure; perception-iteration milestones tune masking and contour filtering; full fix (segmentation) is Phase 2.
Avaota A1 / Limbic board bring-up delays	Medium	Month 1–2 software work can proceed on a dev laptop; board hardening tracked separately and not on the critical path.
Weather / crop-stage availability	Medium	Slack held in weeks 11–16; scope reduces to single-row + single-turn if the window is tight.
1 GbE crossover link on dev laptops	Low	_find_crossover_interface() fallback already handles laptops lacking the end0 interface.

8. Client Q&A

The following addresses questions and comments raised by the APSI / iFarmate team following circulation of the draft plan.

Q1 — Sensor fusion: what is fusioncore and is it already implemented?

Fusioncore is the GNSS/IMU/wheel-odometry fusion node that runs on the Limbic board. It uses a UKF (Unscented Kalman Filter) to fuse three sensor streams:

- **GNSS fix** (`/gnss/fix`) — absolute position from the u-blox F9P rover, with RTK correction when available.
- **Dual-antenna heading** — relative heading from the moving-base/rover F9P pair over the lever-arm baseline; this replaces a magnetometer entirely.
- **Wheel odometry** — incremental displacement from the Lizard motor layer, used to propagate the state estimate between GNSS updates and during dropouts.

The BNO085 IMU (accelerometer + gyroscope, Game Rotation Vector mode) feeds orientation into the fusion as an additional input but is **wired, not yet active** in fusioncore — enabling it is a tracked task in Month 2.

The node is present and running on the current Sowbot hardware. In the Indian suburban environment — where tree cover and buildings may cause multipath or brief outages — the UKF will continue propagating from wheel odometry and IMU when GNSS drops out. ZUPT (Zero-velocity Update) suppresses velocity drift when the robot is stationary, and the Mahalanobis gate rejects noisy fixes. For more detail on the RTK resilience strategy in occluded conditions, see the *Note on RTK GPS in occluded environments* at the end of §4.

Q2 — Hardware: why three F9P receivers?

Three F9P GNSS receivers are required for the heading-without-magnetometer approach chosen for Phase 1:

Role	Count	Purpose
Moving-base (on-robot)	1	Transmits its own raw measurements to the rover via UART; enables the moving-base differential mode.
Rover (on-robot)	1	Receives moving-base data; computes the NAV-RELPOSNEED relative-position vector between the two antennas, giving accurate

Role	Count	Purpose
		heading from the fixed lever-arm baseline (≥ 0.3 m).
Static base station	1	Placed at a known survey point at the field edge; streams RTK corrections (RTCM) to the rover over Wifi link to achieve centimetre-level fix quality.

The two on-robot receivers also provide a single-receiver absolute fix fallback if the moving-base link is lost. The base station is only needed for RTK quality and can be replaced by an NTRIP network subscription (cellular) if local base infrastructure is unavailable — this is one of the items to validate in the Indian field environment (see §4 RTK note).

Q3 — Weekly support and communication: what does that look like in practice?

Sowbot (Agroecology Lab) provides active weekly technical support throughout the engagement. The working pattern is:

- **Weekly sync call (~1 hr) Monday 11am UK time : Saul & Sam usually attend** structured around the open tickets and any blockers. Both developers attend. Agenda circulated beforehand; recordings optional.
- **Async channel (Discord):** day-to-day questions, quick unblocking, and hardware bring-up support. Target response within one working day.
- **GitHub (feldfreund_devkit_ros):** all code review happens here via pull requests. Sowbot reviews and merges each week. The issue tracker (issue #37 status thread) is the canonical progress record.
- **Code review (~2 hr/week Sowbot time):** each merged PR receives written feedback. Larger architectural decisions are discussed before implementation to avoid rework.

If a developer is blocked on hardware or a non-trivial integration question, raise it in the Discord channel first — most issues resolve faster there than waiting for the sync call.

Q4 — Code base and licensing: can the APSI team use and modify the code commercially?

`feldfreund_devkit_ros` (the Sowbot / Feldfreund platform codebase) is **open-source under Apache 2.0**, which permits commercial use, modification, and redistribution with attribution. A verified dependency licence audit covering all runtime and build dependencies is maintained in the repository (https://github.com/Agroecology-Lab/feldfreund_devkit_ros/tree/caatinga-dev#sowbot--feldfreund_devkit_ros--dependency-licence-audit).

The one material restriction is **Forest3D**, the Gazebo simulation package used for maize-field sim worlds. It is licensed **GPL-3.0**, which means it must stay simulation-only and must not be linked into any shipped product binary. All Phase 1 work uses it for sim validation only; nothing in the runtime nav stack depends on it. This boundary is already enforced in the Docker build targets.

Upstream copyright notices for bundled third-party components must be retained as-is — the audit document records which packages carry this requirement. If the iFarmate product eventually ships a derived binary, a licence review pass before release is advisable.

Q5 — Testing options: what can be tested before field access is available?

Not all testing requires a maize field. The three-tier testing ladder used in this plan:

1. Indoor bench / lab (no outdoor access needed)

- Neo perception pipeline: `sowbot_row_follow` can be developed and tuned against a row of real or plastic, potted plants. This is how weeks 3 and 5–6 are structured — no outdoor space required.
- Lizard motor bring-up and teleop: can run on the chassis indoors once the drivetrain retrofit is complete.
- ROS 2 communication and TF graph: fully testable on a single laptop or the SBC pair with no sensors attached.

2. Outdoor (open sky, no crop)

- F9P bring-up and RTK fix quality: needs open sky. A car park or garden is sufficient for weeks 4–5; the dual-antenna heading test only needs ~0.3 m clear baseline, so even a small outdoor area works.

- GNSS fusion stability: the 10-minute static log test in week 5 needs open sky but not a field.
- Full drive test (teleop + E-stop): week 7 integration on any flat outdoor surface.

3. Field (maize crop required)

- First real-row traversal: week 8 — this is the first milestone that genuinely needs standing crop rows.
- All subsequent perception-iteration and repeatability milestones (weeks 11, 13–16) need field access.

The implication for scheduling: the entire Months 1–2 software stack can be built and de-risked before the team needs field access, which helps in environments where plot access is weather- or season-dependent.

9. Action items

From the 5 June 2026 meeting, carried into this plan:

Sowbot side:

1. Draft Phase 1 navigation plan (this document).
2. Prepare and share the Statement of Work based on this plan. Agree a weekly invoicing cadence
3. Share the hardware list above and the hardware request form.

APSI side:

4. Complete the Sowbot installation and setup.
5. Continue the identified small starter tickets.
6. R&D on OpenCV functionality relevant to row detection (feeds Neo work).
7. Explore the F9P RTK GNSS device (feeds Limbic work).
8. Decide the drivetrain route — RoboClaw 2x15A vs brushless ODrive — before week-1 procurement; if ODrive, seek Sowbot advice and add the CAN PAL transceiver.

9. Confirm whether Sowbot should supply the custom ESP32 carrier PCB and clean power supplies, or whether the Indian team already has equivalents.

Tracking: status against milestones to be kept in the live status thread (issue #37) with the weekly merged PRs from both developers as the primary progress signal.